Neurological Regeneration Benefits Associated With The Medicinal Use Of Hericium Erinaceus

(The effects that commonly known, Lion's Mane Mushroom has on treating Alzheimers' Disease)

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INTRODUCTION

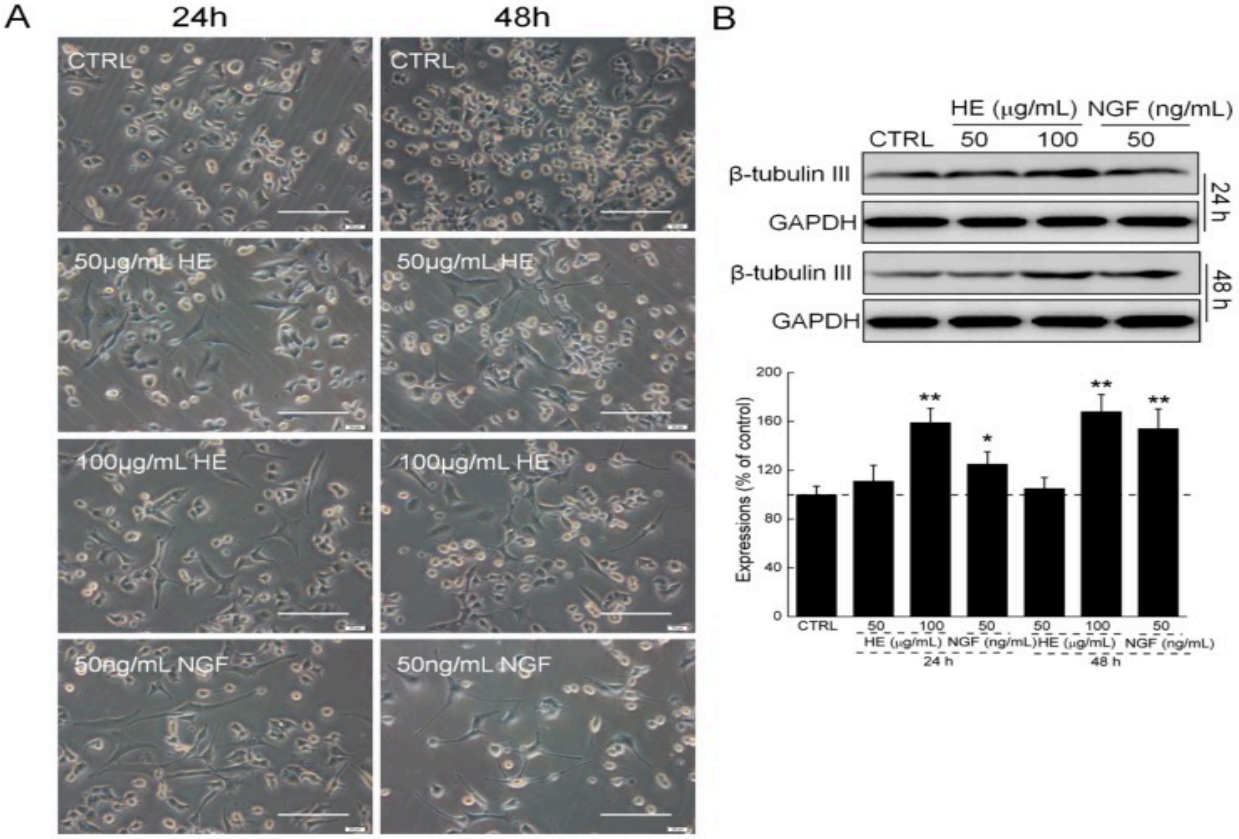
Alzheimers' Disease, what a scary two words. The connotation is of utmost negativity, knowing that the mind will soon enough dwindle into the unconscious world. Alzheimers' Disease is classified by the Alzheimers' Society as a progressive neurological disease in which neuron connections are lost due to protein buildup in the brain, also known as Plaques and Tangles. This disease can develop at any age, but is most commonly developed in people after the age of 65. According to the Centers for Disease Control and Prevention, 5.8 million Americans are currently living with Alzheimers' Disease. (Alzheimers Society, 1). Knowing this, it is quite apparent that action needs to be taken to treat this daunting illness. Because health is the topic area for the Grand Challenges Scholars Program, throughout this reading, three main studies working to treat Alzheimers' Disease will be addressed, one of which will include research from Arizona State University. Each study has its own approach and technique used in order to find a solution for this disease, one will talk about the neuroprotective agents of Hericium Erinaceus, another will talk about the use of focused ultrasound therapy in order to treat current symptoms of the disease, and the final one will discuss potential therapeutic effects of Berberis *darwinii* stem bark in treating Alzheimers' Disease. These studies will all give clear insight into how to further implicate these solutions in the near future to change the lives of millions.

DISCOVERY

Hericium Erinaceus plant (medicinally used Lion's Mane Mushroom) has shown to have various positive effects on the prevention of dementia in patients with Alzheimers' Disease. To begin, in order to understand the context of this study, it will first need to be understood what the function of Choline and Acetylcholine are, as well as how they pertain to the relation of the study. Choline is a chemical produced within the brain that helps to produce acetylcholine, and acetylcholine is a neurotransmitter responsible for the enactment of memory, muscle movement, and mood.(Maternal Choline Supplementation, 1)(This is the ASU study). The present lack of choline and acetylcholine within the brain seem to have a huge impact on the inclining symptoms of Alzheimer's

Disease. In this current study, an Alzheimers' induced rat is used as the subject of testing, and when this Hericium Erinaceus extract was administered to the Alzheimers' rat model, this data then revealed that the Hericium Erinaceus that was used, successfully produced PC12 cell differentiation. Another notable result of this study was that the cholinergic system function was significantly improved and this was shown because the HE dose-dependently enhanced the acetylcholine (Ach) and choline acetyltransferase (ChAT) concentrations in both the serum and the hypothalamus. (Neuroprotective properties, 1). These two results provide experimental evidence that the Hericium Erinaceus may provide neuroprotective agents that can prevent against neurodegenerative diseases. (This is an incredibly remarkable discovery, as this opens a door of the development of new treatments and preventions of a disease that affects millions of people. This is a great indicator that not only could Hericium Erinaceus effect the symptoms and development of Alzhiemers' Disease, but this could become treatment for a multiplicity of neurodegenerative diseases like Parkinson's, Dementia and even Neurological damage.

Figure 1



DIAGRAM

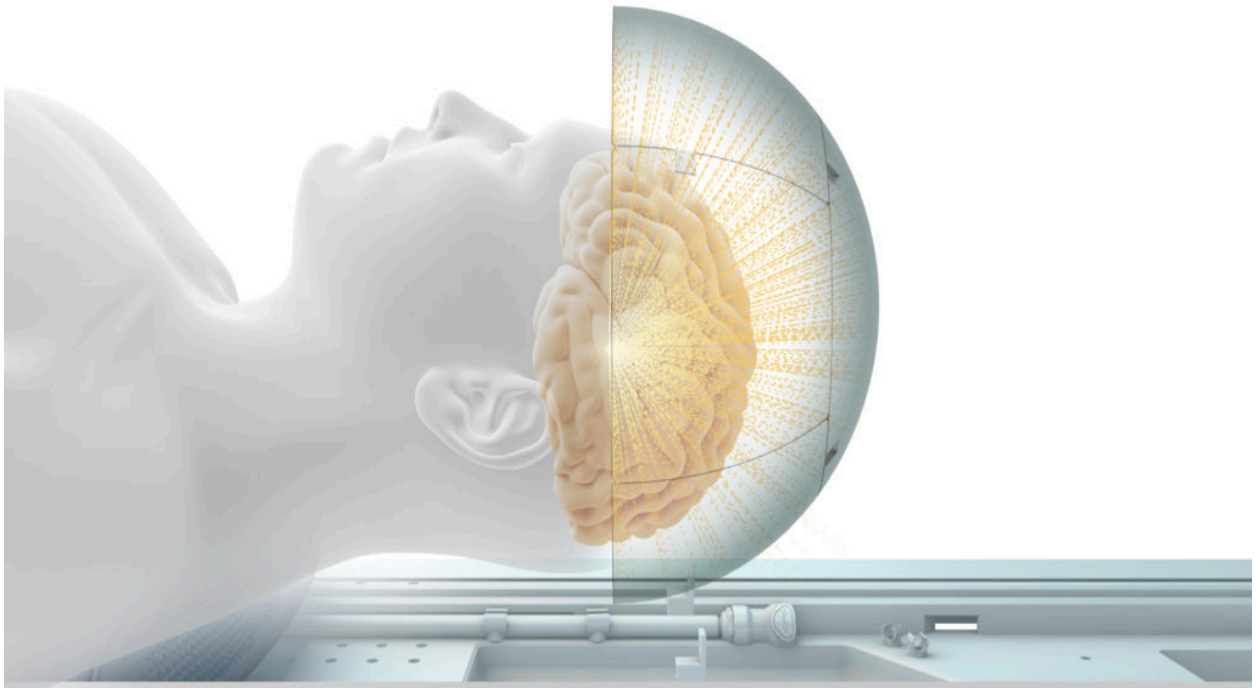
Here it can be seen that from the specimens that were examined, the Hericium Erinaceus has improved cell viability and nuclear apoptotic alternation. It can also be seen that the HE and the NGF applications promoted cell differentiation, and improved the clarity of the neurons in the brain. This proves the discovery made in the previous paragraph.

CONCLUSION

Understanding that this research has proved to have a significant effect on the protection of the neurological processes through the use of the Hericium Erinaceus gives an idea of how this problem can be resolved. This research also provides new information on the successes of natural medicine in recovery of high level disorders and illnesses. By using this information, it is possible that new medicines, research and discoveries can be made within this topic, but, this could also provide affordable help to those in countries without readily available health care because Hericium Erinaceus is so widely available. Although this research is still under heavy experimentation, it is still remarkable to see the discoveries that have already been made on this topic. By doing more research, in which a clinical trial can be developed, there will be better understanding of the true effects that Hericium Erinaceus has on reducing the symptoms of already developed Alzheimer's Disease, and preventing the development of Alzheimer's Disease in at-risk individuals.

Focused Ultrasound Therapy

(To Increase The Quality Of Life In People With Alzheimer's Disease)



INTRODUCTION

In today's world, science is able to develop new and innovative technologies in order to help with all kinds of issues, but, most of these developments are treatments of a current issue rather than a cure or a prevention of the development of that issue. This type of research being done is focused on the treatment of the symptoms of Alzheimers' Disease rather than the prevention or reversal of Alzheimers' Disease like the previous research. This research being done will go over the effects of Focused Ultrasound Therapy in treating patients already diagnosed with AD in order to improve their quality of life. It is important to understand that there are multiple techniques being researched in order to help with the effects of this illness, this one specifically, works to treat the symptoms of a patient that already has been diagnosed with Alzheimer's Disease. With this research, the effects of the disease are not able to be reversed or prevented, but instead this focuses on the use of Ultrasound waves to adhere to the symptoms but also in order to prolong the life and livelihood of the diagnosed individual for as long as possible. Introducing this research will allow us to better understand the ways that this illness can be treated and life can be preserved when faced with this challenge, in order to help create newly developed treatments and maybe eventually cures for this disease.

DISCOVERY

Many discoveries were made in regard to this research, and this is leading to it becoming a highly used practice in the treatment of Alzheimer's Disease in current days. Focused Ultrasound Therapy is the use of Ultrasonic energy specifically focused to a target deep within the brain, without damaging surrounding tissue. This practice is one-hundred percent non-invasive, and allows for a treatment with a short recovery time, and prominent results. In this study, focused ultrasound waves were used to make the blood brain barrier (BBB) more permeable in order to administer medication to the target area intravenously. When this was tested on rat models with Alzheimer's Disease, the rat models were injected intravenously with FUS and MRI imaging agents, as well as anti-Aβ antibodies, BAM-10. (Antibodies Targeted, 1)(Focused Ultrasound Therapy, 1) In order to increase the permeability of the BBB, before they were given the injections, they were first administered focused ultrasound therapy. Within minutes of the injection, it was already visible that the MRI imaging agent had entered the brain, and later they would find that the BAM-10 antibody was bound to the plaques in the target area. This shows us that the use of focused

ultrasound therapy significantly increased the permeability of the BBB so that it would be possible to administer treatment intravenously, while also ensuring that it will reach the target area within the brain. This is an extremely important discovery because this helps us come to the conclusion that by using this focused ultrasound therapy, we would be able to administer so many treatments to the brain that have never before been able to reach a targeted area in the brain. This will allow for many new treatments for Alzheimer's Disease, but it can also lead to the discovery of treatment for so many other neurological illnesses, and diseases.

CONCLUSION

Alzheimer's Disease can leave friends and family feeling lost, and grieving for much longer than the effects of other losses. Families start to grieve as soon as the dementia begins to set in. The slow decline of neurological function takes its toll for much too long. With this research, there seems to be a light at the end of that long grieving tunnel. By allowing doctors to administer treatments directly to the target in the brain, rather than just administering it to the bloodstream, treatments for these cognitive illnesses becomes increasingly easier, and more effective. With efficacy declining, this can allow for people diagnosed with this disease to have more of the good years to spend with their loved ones. This research could become a leading step in the development of new treatments for Alzheimer's Disease, this could be the next future in neuroscience, and medicine.

RELATION TO GRAND CHALLENGES SCHOLARS PROGRAM

These new studies being conducted are also very relevant to the Grand Challenges Scholars Program. Given that the program is all about innovation, and developing new ways to solve very prominent issues in society, this research falls in line with that program perfectly. Throughout the time in the program, this research can be further developed and provide insight into the future of medicine. It is important for students in the program to pay attention to new and improving research and discoveries, because soon that research will be applied to the work that we are doing in the near future.

Anti Tau DMT's

(Prevention of tau-Protein Phosphorylation or Fibrillarization, and Promotion of Clearance)

INTRODUCTION:

The final research that will be presented is still in early stages, and has little discovery in the area, but could lead to many future discoveries for Alzheimer's Disease. This research has to do with preventing the onset of AD, by preventing the production of the Tau protein, which is the main cause of neuron degeneration. This research is extremely relevant to the Grand Challenges Scholars Program, because it is still under a ton of research and is more focused toward creating a future solution, which is what Grand Challenges is all about. This research not only will help make discoveries on the Tau proteins and their effects on AD, but, it will also lead to developing new innovative preventions for AD, and maybe even help to find a complete cure for the disease.

DISCOVERY:

This next research that will be addressed is research specifically on disease modifying treatments, or DMTs. The research that will be the main focus here is what they are doing to create a treatment to fight the production of Tau proteins which are one of the main causes of neuron dysfunction. Anti-phospho-tau approaches are still fairly new and have not even reached stage III of development. Potential ways that are currently being studied in order to prevent tau phosphorylation and fibrillization are glycogen synthetase kinase-3 (GSK-3) inhibitors (e.g., tideglusib–did not meet phase II endpoints), τ assembly inhibitors and aggregation inhibitors (e.g., methylene blue[113] and its derivative TRx0237), and microtubule stabilizers (e.g., davunetide–did not meet week 12 phase II endpoint) (West, 1)(ANTI-Tau DMTs, 1).There is also the fairly new development of anti-tau passive immunotherapies. These all work to lessen the production of tau proteins in order to prevent and delay the onset of Alzheimer's Disease.

Figure 2:

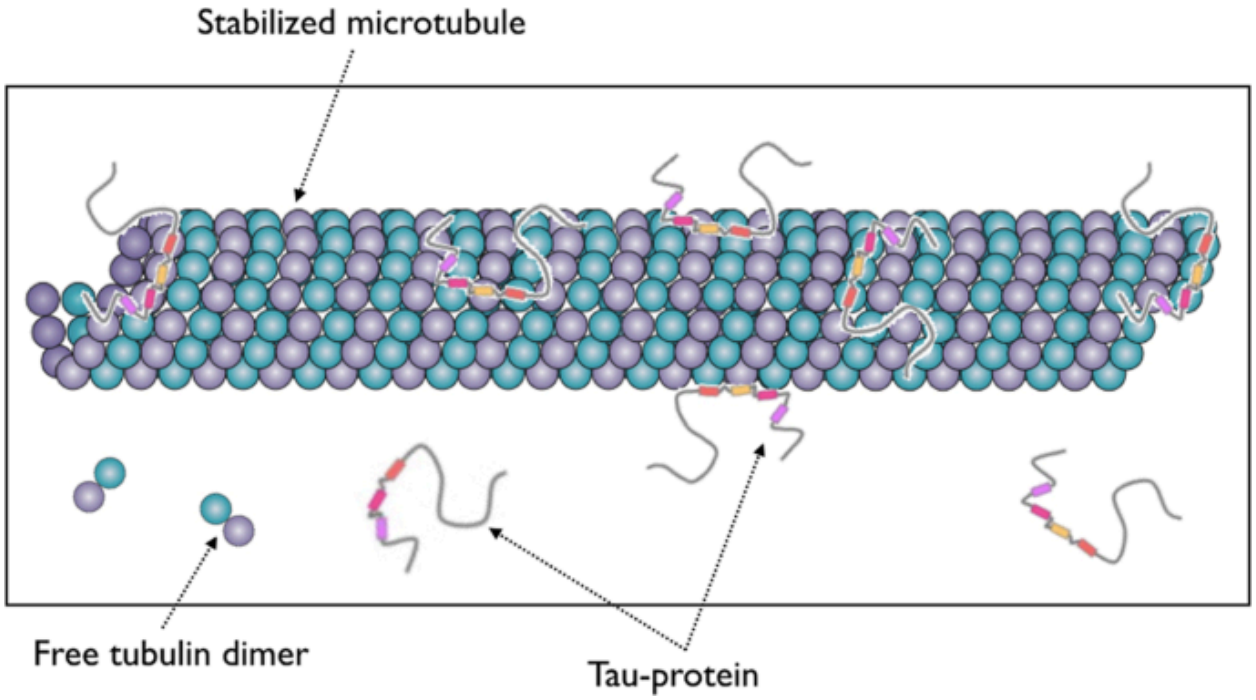


Figure 3:

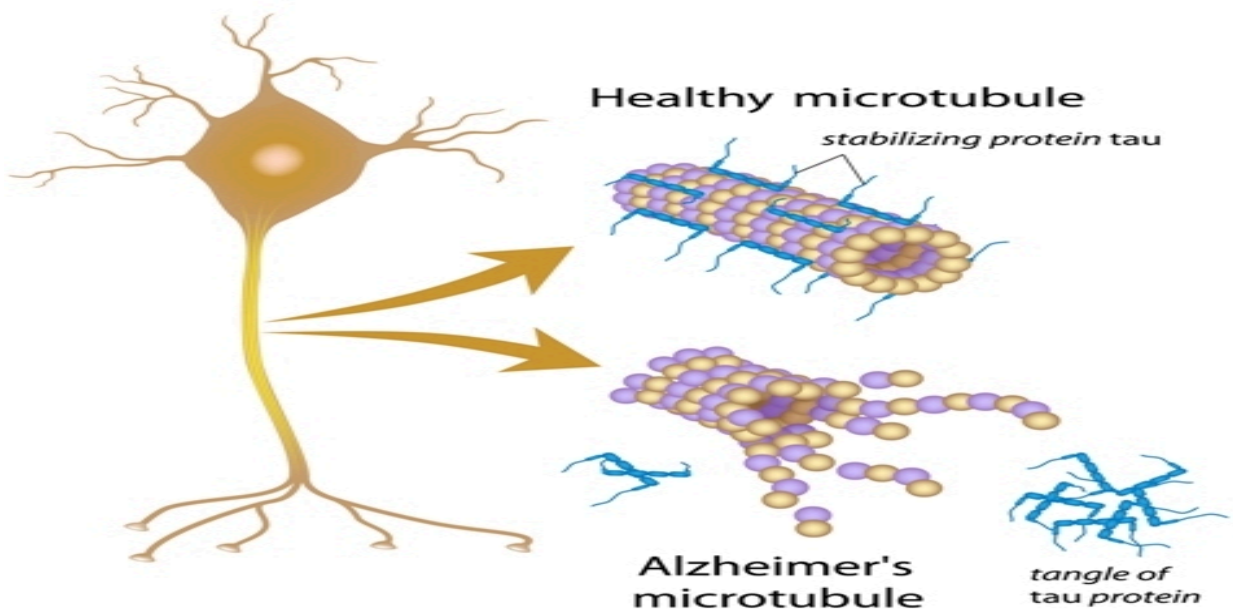


DIAGRAM:

As can be observed from the picture above, it is clear that when the tau protein latches on to the stabilized microtubules, it is tearing off some of the tubulin dimers, which creates free tubulin dimers. Then from the second picture it can be seen that when these tubulin dimers begin to detach from the microtubules, the microtubules begin to unhinge from itself, creating a destabilized microtubule, which resembles what the microtubules would look like in a patient with Alzheimer's Disease. The research that this is addressing is working to prevent the tau proteins from being created in order to stop this process of destabilization from happening.

CONCLUSION:

In conclusion, there is a multiplicity of research being done every single day to work towards the treatment, prevention or cure of Alzheimer's Disease. This terrible, painful, mentally degenerative disease ruins the lives of so many. Through this research, scientists are finding new ways towards the forefront of neurodegenerative treatments. Discoveries thus far prove to be extremely remarkable, and extremely relevant to the subject area of health in the Grand Challenges Scholars Program. These studies will be the future of Alzheimer's medicine, and can be implemented into many future projects focused towards health for the Grand Challenges Scholars Program.

Work Cited

"Alzheimer's Disease." *Alzheimer's Society*, 30 July 2021, www.alzheimers.org.uk/about-dementia/types-dementia/alzheimers-disease.

"Alzheimer's Disease." *Alzheimer's Society*, 30 Sept. 2021, https://www.alzheimers.org.uk/about-dementia/types-dementia/alzheimers-disease.

"Alzheimer's Disease." *Focused Ultrasound Foundation*, https://www.fusfoundation.org/diseases-and-conditions/neurological/alzheimers-disease#:~:text=Focused%20 ultrasound%20is%20an%20 early,for%20patients%20with%20Alzheimer's%20disease.

Atri, Alireza. "Current and Future Treatments in Alzheimer's Disease." *Seminars in Neurology*, Thieme Medical Publishers, 29 Mar. 2019, https://www.thieme-connect.com/products/ejournals/html/10.1055/s-0039-1678581#N71323.

Jordão, Jessica F., et al. "Antibodies Targeted to the Brain with Image-Guided Focused Ultrasound Reduces Amyloid-β Plaque Load in the TgCRND8 Mouse Model of Alzheimer's Disease." *PLOS ONE*, Public Library of Science, https://journals.plos.org/plosone/article?id=10.1371%2Fjournal.pone.0010549.

"Preclinical and Clinical Development of ABBV-8E12, a Humanized Anti-Tau Antibody, for Treatment of Alzheimer's Disease and Other Tauopathies." *PRECLINICAL AND CLINICAL DEVELOPMENT OF ABBV-8E12, A HUMANIZED ANTI-TAU ANTIBODY, FOR TREATMENT OF ALZHEIMER'S DISEASE AND OTHER TAUOPATHIES • The Journal of Prevention of Alzheimer's Disease*, http://www.jpreventionalzheimer.com/2825-preclinical-and-clinical-development-of-abbv-8e12-a-humanized-anti-tau-antibody-for-treatment-of-alzheimers-disease-and-other-tauopathies.html.

Velazquez, Ramon et al. “Choline as a prevention for Alzheimer's disease.” *Aging* vol. 12,3 (2020): 2026-2027. doi:10.18632/aging.102849

“What Is Alzheimer's Disease?” *Centers for Disease Control and Prevention*, Centers for Disease Control and Prevention, 26 Oct. 2020, www.cdc.gov/aging/aginginfo/alzheimers.htm.

Zhang, Hua, et al. “Store-Operated Calcium Channel Complex in Postsynaptic Spines: A New Therapeutic Target for Alzheimer's Disease Treatment.” *Journal of Neuroscience*, Society for Neuroscience, 23 Nov. 2016, www.jneurosci.org/content/36/47/11837?utm_source=TrendMD&utm_medium=cpc&utm_campaign=JNeurosci_TrendMD_0.

Zhang, Junrong et al. “The Neuroprotective Properties of Hericium erinaceus in Glutamate-Damaged Differentiated PC12 Cells and an Alzheimer's Disease Mouse Model.” *International journal of molecular sciences* vol. 17,11 1810. 1 Nov. 2016, doi:10.3390/ijms17111810